



Asset Performance:

Pol 05 Refrigerant leak detection systems



Credits



No Minimum Standard

Aim

To reduce the level of greenhouse gas emissions related to the leakage of refrigerants.

Question

Is there an automated refrigerant leak detection system in place for all equipment that uses refrigerants?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No leak detection system in place
3	C.	Yes, warning alarm or lighting only
4	D.	Yes, warning alarm or lighting, automatic shutdown and pump down of refrigerants
4	E.	Only environmentally benign refrigerants (GWP ≤ 1 , or solid-state) are used

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where no refrigerants are used or only small (refrigerant charge in each system is $\leq 5\text{kg}$) hermetic systems are installed in the asset, this issue can be filtered out of the assessment.	All
2.	A leak detection system should be in place for systems that are installed in the building for the following uses, including, but not limited to: a) Walk-in cold storage enclosures b) Cold storage, including commercial food/drink display cabinets but excluding residential-scale white goods (e.g. fridges and freezers) c) Comfort cooling and heating (e.g. heat pumps) d) Process based cooling loads (e.g. servers/IT equipment)	All

Criterion	Assessment criteria	Applicable Answer
3.	A list of typical refrigerants with a low GWP can be found in Table 27 in the Checklist and Tables section of Pol 04 Global warming potential of refrigerants	E
4.	This issue applies to refrigerants used in equipment that is installed on-site only.	All

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Operation and maintenance manuals detailing installation of leak detection system or site inspection
All	In the case of environmentally benign refrigerants, manufacturer's confirmation of either minimal risk of leaks or minimal impact of leaks.
1	Statement from the building manager indicating that the asset does not contain any systems that contain refrigerants or confirmation that the total charge is $\leq 5\text{kg}$ in any systems that are present.